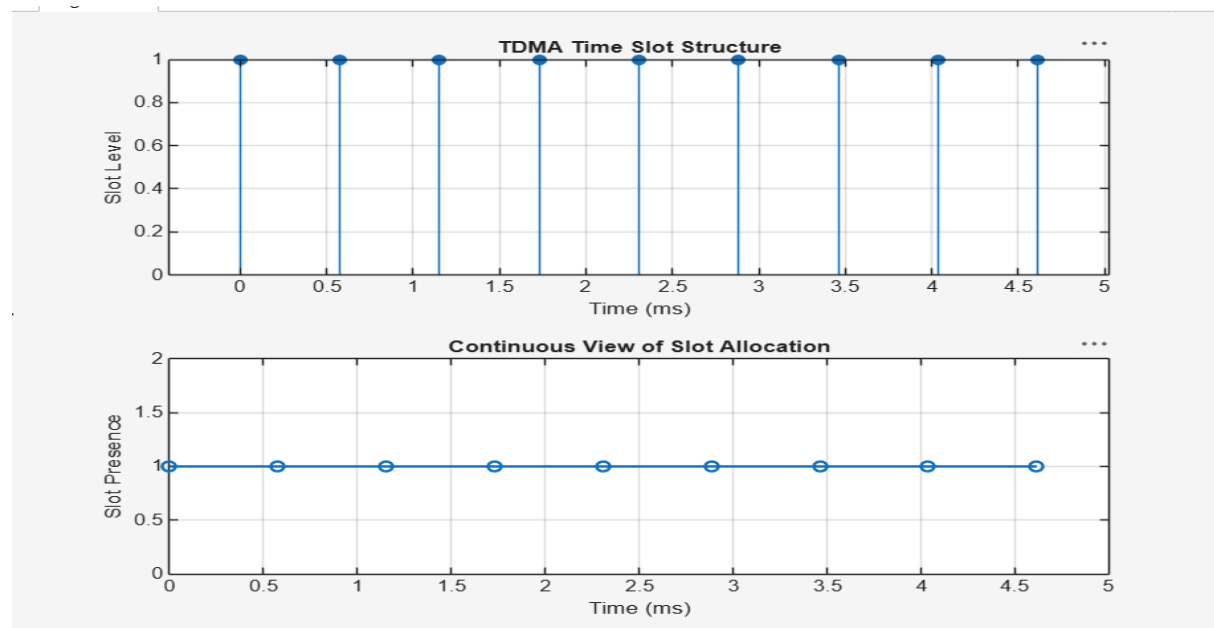


EXP 2

Objective 1

```
clc; clear; close all;  
FrameTime = 4.615; Slots = 8;  
SlotDuration = FrameTime/Slots;  
t = 0:SlotDuration:FrameTime;  
subplot(2,1,1)  
stem(t,ones(size(t)),'filled')  
title('TDMA Time Slot Structure')  
xlabel('Time (ms)')  
ylabel('Slot Level')  
grid on  
subplot(2,1,2)  
plot(t,ones(size(t)),'-o','LineWidth',1.5)  
title('Continuous View of Slot Allocation')  
xlabel('Time (ms)')  
ylabel('Slot Presence')  
grid on
```

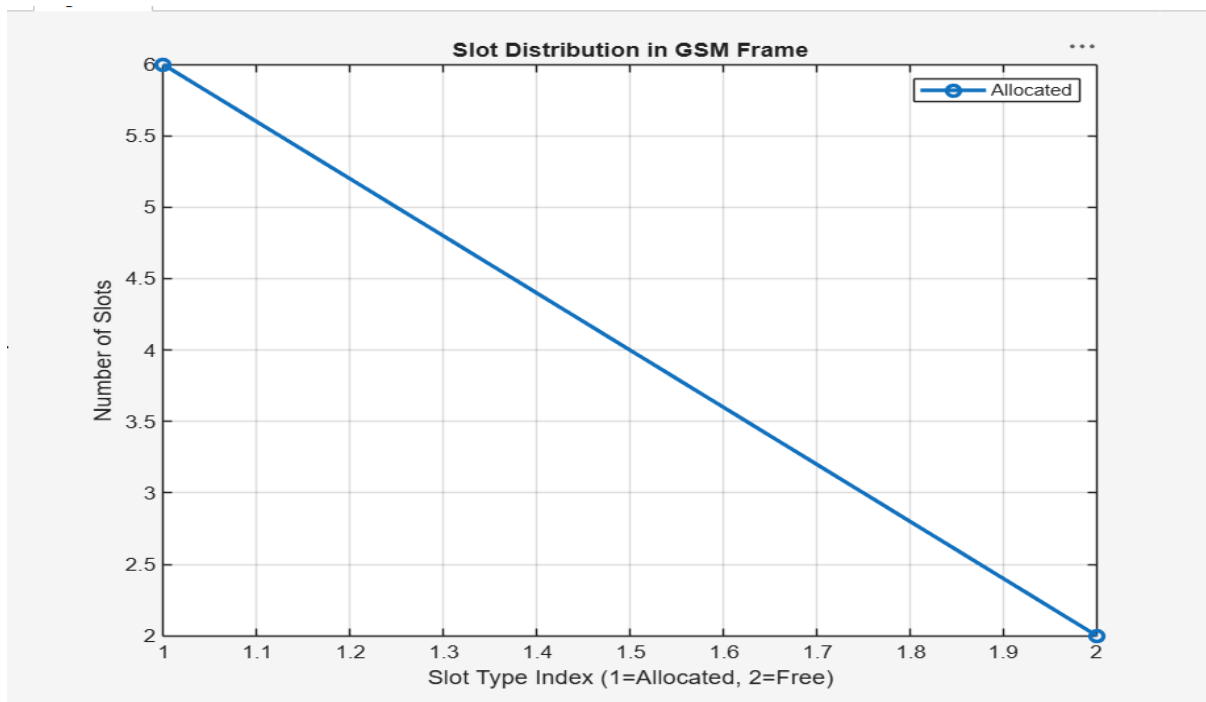
OUTPUT



Objective 2

```
clc; clear; close all; TotalSlots =8;
AllocatedSlots = 6;
FreeSlots = TotalSlots -
AllocatedSlots;
Utilization =
(AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot
(SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots
FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization
(Allocated vs Free)')
xlabel('Slot Type Index
(1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on
% Graph 2: Pie Chart (SAFE)
subplot(2,1,2) pie([AllocatedSlots
FreeSlots])
title('Slot Distribution in GSM
Frame')
legend('Allocated','Free')
```

OUTPUT



Objective 3

```
clc; clear; close all;
Users = 1:8;      % Total users
TotalSlots = 5;
Alloc = zeros(1,8); % Initialize
allocation
Alloc(1:TotalSlots) = 1;
Blocked = 1 - Alloc;
disp('User Wise Slot Allocation')
disp(table(Users', Alloc', ...
    'VariableNames',
    {'User','Status_1_Allocated_0_Bl
    ocked'}))
% Single Figure with Two
Graphs
figure;
% Graph 1: Allocation per User
subplot(2,1,1)
```

```

stem(Users, Alloc, 'filled',
'LineWidth', 2)
title('TDMA Slot Allocation per
User')
xlabel('User Index')
ylabel('Allocated')
ylim([-0.2 1.2])
grid on
% Graph 2: Slot Utilization
subplot(2,1,2)
plot(Users, Alloc, '-o',
'LineWidth', 2)
hold on
plot(Users, Blocked, '-s',
'LineWidth', 2)
title('TDMA Slot Utilization
Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked','Loca
tion','best')
ylim([-0.2 1.2])
grid on

```

Output

